

Measuring selection on reaction norms: Lack's principle and plasticity in clutch size

Jordan S. Martin^{1,2}, David F. Westneat³, Shinichi Nakagawa⁴, Julia Schroeder⁵, Terry Burke⁶

¹Department of Fish Ecology and Evolution, Eawag, Kastanienbaum, Switzerland

²Institute of Evolutionary Medicine, University of Zurich, Zurich, Switzerland

³Department of Biology, University of Kentucky, Lexington, Kentucky, United States

⁴School of Biological, Earth and Environmental Sciences, University of New South Wales, Sydney, Australia

⁵Department of Life Sciences, Imperial College London, South Kensington Campus London, United Kingdom

⁶School of Biological Sciences, University of Sheffield, Western Bank, Sheffield, United Kingdom

Corresponding author: Department of Biology, University of Kentucky, 101 Morgan Building, Lexington, KY 40506-0225, United States.

Email: david.westneat@uky.edu

J.S.M. and D.F.W. contributed equally.

Present address: S.N. is now at the Department of Biological Sciences, University of Alberta, Edmonton, Alberta, CA.

Abstract

Plastic phenotypes may often be subject to conflicting demands, which should generate nonlinear selection favoring intermediate optima. However, investigating complex patterns of selection on reaction norms has been challenging. We leveraged data on clutch size from 2 long-term studies (25 and 19 years) of individually marked house sparrows (*Passer domesticus*). We used a novel multivariate technique for examining linear and nonlinear selection acting on clutch size reaction norms via the fitness components of hatchling number, nestling survival, and nestling body mass at the end of the main parental period. Reaction norm slopes were highly canalized and lacked sufficient among-female variation to detect selection. Stabilizing selection and opposing patterns of directional selection occurred on both intercepts and individual residual variation via hatchling counts and nestling body mass. We also observed changes in these gradients as a function of laying date. Our results support Lack's hypothesis that quantity–quality tradeoffs shape selection on both mean clutch size and the variability of clutch sizes across breeding seasons. Our findings are also consistent with models of adaptation on asymmetric fitness landscapes, where tradeoffs near a local fitness ridge or cliff favor distinct patterns of clutch size plasticity in response to unpredictable environmental variability.

Keywords: phenotypic plasticity, stabilizing selection, tradeoffs, canalization, residual variation

Introduction

Phenotypic plasticity, the variance in phenotype within a genotype due to variance in the environment (Pigliucci, 2001), is ubiquitous among living organisms. For many traits in many types of environments, this variation can be described by a reaction norm (Woltereck, 1909)—a function, potentially of any shape, relating phenotype to the environment. If plasticity is adaptive, such as in response to environmental fluctuations in optimal phenotypes (de Villemereuil et al., 2020), then several parameters of a reaction norm may be under selection and could evolve. The ecological basis of such selection is understood in broad outline (e.g., Botero et al., 2015; Moran, 1992; Scheiner, 1993, 2013; Scheiner & Holt, 2012), but further details are critical to understanding the specific ways plasticity manifests in and shapes the evolution of particular traits and species.

Function-valued traits (Pletcher & Geyer, 1999) are those that exhibit plasticity in response to one or more continuous environmental factors and may best be conceptualized as a continuous function (infinite-dimensional) rather than a single trait value (Kirkpatrick & Heckman, 1989). Repeated measures of function-valued traits thus can be used to quantify an individual's phenotype as a curve across en-

vironments, allowing for the curve itself, i.e., the reaction norm, to be treated as a biologically meaningful target of natural selection and evolution (de Jong, 1995; Kingsolver et al., 2014). Various kinds of functional analysis, ranging from standard additive linear models to more complex, non-parametric methods such as basis function expansion, have been developed for the empirical study of reaction norms (Gomulkiewicz et al., 2018). Random regression, which can flexibly incorporate both linear and nonlinear curvature, has emerged as the most popular and commonly used parametric method for quantifying reaction norms and estimating individual (co)variation in the parameters composing these functions (Dingemans et al., 2010; Meyer & Kirkpatrick, 2005; Stinchcombe et al., 2012).

While some function-valued traits such as thermal performance are characteristically nonlinear, linear reaction norms often provide useful approximations for the curvature of traits such as reproductive behavior with respect to the local range of environmental variation, under appropriate transformation of the data scale, for which generalized random regression methods are well suited (Dingemans & Doctermann, 2013). A strictly linear reaction norm has at least three attributes of relevance: the intercept, a slope in

response to a measured environment, and the residual deviations of each expression of the trait from the target reaction norm. Each of these could shape fitness and, hence, experience selection. Despite considerable theory on the conditions influencing the evolution of plasticity (Botero et al., 2015; Dewitt et al., 1998; Moran, 1992; Scheiner & Holt, 2012), as well as the importance of plasticity for explaining evolutionary persistence and change in variable environments (e.g., Chevin et al., 2010; Gavrillets & Scheiner, 1993), empirical measurement of selection on reaction norms has been limited and produced few clear relationships (Arnold et al., 2019; Kingsolver et al., 2007; Pigliucci & Schlichting, 1996; Weis & Gorman, 1990). Understanding how selection is actively shaping the reaction norms of organisms in natural populations has been challenging, due to both the potential complexity of reaction norms as well as the selection acting on them.

Making progress on this challenge is complicated by statistical issues. Arnold et al. (2019), for example, found that only eight studies had directly tested for selection on thermal reaction norm slopes despite the large research effort devoted to explaining adaptive plasticity in response to temperature. Some previous studies have estimated individual slopes, treated these as a measurable phenotype, and conducted secondary analyses in which fitness metrics are regressed on the extracted measures (e.g., Brommer et al., 2005; Charmantier et al., 2008; Nussey et al., 2005). This approach is compelling, as it allows nonlinear selection to be estimated, and graphical representations of the data allow one to envision the selection surface and local adaptive landscape. However, doing such secondary analyses has come under criticism (e.g., Hadfield et al., 2010; Houslay & Wilson, 2017) because reaction norm attributes are estimates with imprecision, and are calculated from varying amounts of information, exhibit varying levels of error, and so arise from a family of distributions whose underlying variance is not precisely known. For some questions, this is unlikely to affect estimates of the magnitude of selection, but it will affect the confidence of those estimates and, thus, any rigorous assessment of their statistical support and biological importance, leading to anticonservative inference (Hadfield et al., 2010). This is particularly problematic for investigating selection on reaction norm slopes and residuals, which may often experience weaker but covarying selection with reaction norm intercepts and thus tend to be inferred with greater uncertainty and imprecision (Auld et al., 2010).

One solution to this problem is to treat both the trait and fitness as dependent variables (e.g., Arnold et al., 2019; Brommer et al., 2012; Dingemanse et al., 2021; Hadfield et al., 2010; Morrissey et al., 2012) in a bivariate random regression model (Meyer, 1998), which simultaneously accounts for uncertainty in phenotypic reaction norms and their fitness effects by directly estimating the covariance between these reaction norm parameters and the fitness components of interest. These random effect covariances can then be converted into and interpreted as selection gradients, providing estimates unbiased by stochastic causes of phenotypic variation such as measurement error and spatiotemporal heterogeneity (Dingemanse et al., 2021; Stinchcombe et al., 2014). Related approaches have also been proposed to extend methods for estimating function-valued traits to measure continuously varying directional selection gradient functions (Gomulkiewicz et al., 2018).

The covariance between trait and fitness provides an estimate of directional selection (Price, 1970). However, a reasonable hypothesis about plasticity is that most reaction norms seen in nature are under some form of stabilizing selection, which is caused by a nonlinear relationship between trait and fitness and thus cannot be measured by linear covariance functions. There are multiple reasons to expect stabilizing selection. First and foremost, the phenotypes produced by plastic mechanisms likely experience tradeoffs, favoring an intermediate optimum (Lack, 1947; Smith & Fretwell, 1974; Williams, 1966). That optimum could vary with environmental conditions, thereby favoring a reaction norm that targets the optimal trait value in different environments (de Villemereuil et al., 2020; Scheiner, 1993; Tufto, 2015). One potential solution employing the standard multivariate approach is to split fitness into the relevant components of the tradeoff and test whether their covariances with traits have opposite signs. This is a crude fix, however, as most models of tradeoffs rely on at least one component of fitness having a nonlinear relationship with the trait. Other methods for quantifying nonlinear relationships between fitness and phenotype (e.g., Morrissey & Sakrejda, 2013) do not inform as to whether individuals experience differential fitness as a function of the reaction norm parameters producing observed trait values.

Reaction norms with steeper or shallower slopes than the most adaptive ones should produce phenotypes that miss the optimum in either direction and, hence, have lower fitness. In addition, if the optimal reaction norm lies on a fitness ridge, residual deviations in phenotypic expression may reduce fitness by producing suboptimal trait values on the downward slopes surrounding the peak. In that case, selection should favor minimal deviations (i.e., canalization of the phenotype, Flatt, 2005; Wagner et al., 1997; also known as improved predictability, O'Dea et al., 2022). However, reducing such deviations may entail its own unique cost (e.g., DeWitt et al., 1998; Hansen et al., 2006; Westneat et al., 2019), and increasing deviations may in turn result in unique benefits (e.g., Borenstein et al., 2008; Hugie, 2003; Moore et al., 2017), so that some degree of residual deviation may be optimal due to these tradeoffs. However, those tradeoffs coupled with an asymmetric fitness surface may favor suboptimal mean phenotypes (Vercken et al., 2012). To our knowledge, despite widespread evidence of among-individual differences in residual variance (Mitchell et al., 2021) and theory on how selection could act via residual variance (Gavrillets & Hastings, 1994; Hill & Zhang, 2004), no studies have directly assessed selection on residual variance in a natural population. Finally, in some cases of plasticity (e.g., Dingemanse et al., 2012; Westneat et al., 2013), intercept, slope, and residuals can covary across individuals. Such covariance could have a variety of causes at different levels of analysis (Dubois, 2019; O'Dea et al., 2022), with one being the possibility of correlational selection (certain combinations of intercept, slope, and/or residual deviations) driving reaction norm shape. Indeed, stabilizing selection resulting from tradeoffs between reaction norm parameters near fitness ridges should be expected to shape both the intercepts determining expected phenotypes and the slopes and residuals determining variation around the mean (Gavrillets & Hastings, 1994; McGlothlin, 2010; Wagner et al., 1997; Zhang & Hill, 2005). To our knowledge, this latter possibility of correlational selection on reaction norms has

never been tested. None of these issues can be straightforwardly assessed with the traditional bivariate random effects approach introduced above (Dingemanse & Dochtermann, 2013).

Martin et al. (2025) developed a statistical method that extends random regression analysis of plastic traits to allow statistically appropriate measurement of nonlinear selection on reaction norms. Here, we apply this method to study stabilizing selection and tradeoffs shaping the reaction norms of clutch size in birds. Clutch size provides an incisive example of the potential forces affecting plasticity. In 1947, David Lack proposed that clutch size in birds is subject to optimizing selection due to a tradeoff between the number of offspring and their quality. Laying more eggs increases the number of offspring, but because subsequent parental resources (provisioning of food in birds) might be limited and must be distributed among more individuals, having more offspring decreases offspring quality. The form of these tradeoffs, labeled as Lack's principle (Williams, 1966), has received considerable empirical support (Godfray et al., 1991), although other factors also appear to influence the evolution of clutch size (e.g., Pettifor et al., 1988; Rowe et al., 1994; van der Werf, 1992).

Clutch size exhibits phenotypic plasticity, primarily in response to variation in date when the clutch is laid (Brommer et al., 2008; Westneat et al., 2009, 2014) but also to other factors such as territory quality (Högestedt, 1980) and density (Nicolaus et al., 2013). A reasonable hypothesis is that such plasticity is adaptive, in which case there is likely to be stabilizing selection on the attributes of the clutch size reaction norm. Lack did not include plasticity in his ideas about tradeoffs shaping mean clutch size, but his principle could potentially apply to the selection acting on the intercept, the slope, and the fit of a given clutch size to the expected reaction norm (residual).

Here, we use Lack's hypothesis as a focus for measuring (non)linear selection on clutch size reaction norms, providing an instructive example of how such selection can be measured and may reveal novel insights into the key factors proposed by Lack. We applied Martin et al.'s (2025) approach to two datasets of clutch sizes from individually marked female house sparrows (*Passer domesticus*). House sparrows lay two to eight eggs in one to six nests per season, sometimes over several breeding seasons (Anderson, 2006; Westneat et al., 2009). Plasticity in clutch size occurs with respect to date in the season and attempt order, and the two variables interact (Westneat et al., 2009) to produce a warped reaction norm plane in multidimensional environmental space. Westneat et al. (2014) found that populations differed in intercepts, mean slope with respect to date, and in the among-individual variance in intercepts. One hypothesis for these differences is that different locales induce different selection pressures on reaction norm attributes. If Lack's hypothesis is sufficient to explain this, then we predict that in either population studied here, a positive deviation in an individual's intercept from the population mean intercept or a positive deviation in slope from the population mean slope would increase the individual's number of hatchlings (a positive covariance) but would reduce either its nestlings' survival to fledging or their mass at fledging (negative covariance). Moreover, we predicted that a larger residual variation in clutch size (e.g., more deviations from an individual's reaction norm) compared to the population

mean residual variation would result in lower net fitness due to either smaller broods (negative deviations) or reduced offspring quality (positive deviations). If selection is not stabilizing on the population mean, then Lack's hypothesis is not sufficient, and one or more of several alternative processes may be exerting influence either on selection or on the evolution of the reaction norm. If, for example, a time horizon drives selection on clutch size response to lay date (e.g., Rowe et al., 1994), then we might expect our analysis to reveal that across the population, late-season nests deviate from the Lack optimum much more than early-season nests.

Methods

We present analyses of data obtained from two populations of house sparrows, one located in Lexington, Kentucky, USA (KY), and one on Lundy Island, UK (LY). Data collection is described in detail in Westneat et al. (2014), with the addition here of several more years of data (1992 up through 2017 for KY and 2000–2018 for LY). Briefly, marked individuals nesting in boxes erected for monitoring sparrow breeding were censused in each season. We determined clutch size from inspection of nest contents. Clutches start with the laying of an egg in a previously empty nest, a single egg is typically laid every day, and the clutch is complete when laying stops and the female is incubating. Various inefficiencies in fieldwork (such as gaps in surveying and miscounts) and unusual events for the birds (an initial clutch partially destroyed with relaying occurring on top of what remained) sometimes meant that clutch size was difficult to discern. Details on the rules we used in these events can be found in Westneat et al. (2009, 2014) and Cleasby et al. (2010).

We counted the minimal number of eggs that hatched (see Stewart & Westneat, 2013). Nests were not checked often enough to assess for sure if every egg hatched. The minimal hatch was thus the maximum number of nestlings observed to be in the nest after hatching began. If a nest had five eggs, but later only four nestlings and no eggs were found, we assumed only four eggs hatched, and the other was removed either by the parents or a predator. While it is possible in this case that the other egg hatched and the nestling was removed, we used a conservative estimate of initial brood size. On days 5–12 of nestling age, and typically on days 10–12, any nestlings in the nest were banded with a numbered aluminum leg band and weighed. Variation in nestling age at weighing had no discernable effect on females' expected nestling mass. We used the proportion of hatchlings that survived to be banded and the individual mass when banded as metrics of offspring quality. While recruitment of offspring to the breeding population is the ultimate measure of success (Alif et al., 2022), there are two problems with this metric. First, it can be more easily measured on Lundy Island (a closed island population) than for the Kentucky population (a mainland one where dispersal off the study site is common), making it likely that any differences in selection are in fact due to differences in accuracy of the fitness measure. Second, the recruitment probability conflates direct and indirect (parental) effects on fitness (Wolf & Wade, 2001). Measures of success within the nest are likely to be influenced mostly by parental traits, whereas recruitment will be heavily influenced by offspring traits. We thus focused our measures of

fitness on metrics taken while the nestlings were still completely dependent on their parents. We also note that total selection on reaction norms may include other components of fitness, such as carryover effects of one breeding attempt on other attempts by the same adult individual (e.g., effects on residual reproductive values, Williams, 1966). Our initial goal was to investigate whether Lack's hypothesis in a variable ecology was sufficient to explain reaction norm shape, and so here we confine fitness metrics to those measured within each breeding attempt.

Multivariate analysis of selection

We employed the novel method developed by Martin et al. (2025) for estimating nonlinear selection on clutch size reaction norms. This method makes use of Bayesian multivariate/multi-response generalized mixed-effects models (GLMMs) to simultaneously estimate reaction norm parameters and their potentially nonlinear effects on individual fitness. The major advantage of this reaction norm (non)linear selection model is that it can estimate both linear and nonlinear selection acting on multiple elements of reaction norms and components of fitness in a single analysis, using any statistical distributions of interest, while appropriately accounting for uncertainty in both fitness and phenotype in the process (Hadfield et al., 2010). Posterior distributions from this Bayesian model can then be further processed to derive selection gradients appropriate for hypothesis testing of evolutionary ecological theory, carrying statistical uncertainty forward across each stage of analysis (Stinchcombe et al., 2014).

To prepare our data for this reaction norm (non)linear selection model, we started with all nesting attempts of known (banded) females in which clutch size, date in the season, and subsequent fate were known. For Kentucky, this produced 1,771 clutches from 608 females having 1–20 clutches each. For Lundy Island, we tallied 967 clutches from 385 females having 1–12 clutches each. We then tallied entries for each nestling that was banded ($N = 5,907$ for KY, $N = 2,486$ for LY) and weighed along with the clutch size of that attempt, the minimal number of hatchlings, the nestling's mass, the date when the first egg was laid for that attempt, the attempt order in the season for that female, and identifiers for the female and the attempt. We omitted any attempt either with an unmarked female or that did not produce at least one offspring that was banded. We mean-centered date in the season within each female and adjusted attempt order to the number of prior attempts, such that attempt 1 was coded as 0. Clutch size was then modeled as a function of these variables and their interaction.

Clutch size

Previous analyses of house sparrow clutch size (Westneat et al., 2009, 2014) employed Gaussian linear random regression models. Here, we instead used ordered logit random regression models designed for ordinal data to more appropriately account for the fact that clutch sizes were highly underdispersed compared to the predictions of Gaussian or Poisson models. Figure 1 visualizes this pattern, showing how posterior predictions from these models failed to describe the distribution of our data, while the cumulative logit model accurately predicted the sharp peak of observed clutch sizes around four to six eggs. This so-called posterior

predictive check is a crucial step of model validation during Bayesian data analysis (Gelman et al., 2020). The ordinal model performed better in predicting our data because it allowed for varying cumulative probability thresholds between egg counts, such that the probability of a female going from, e.g., three to four eggs was not assumed to be the same as her probability of going from four to five eggs (see Bürkner & Vuorre, 2019 for an accessible formal overview of the ordered logit model). The underlying process in the ordinal model also better fits the presumed biology of clutch size due to (1) eggs being discrete events and (2) females shifting the probability of laying a subsequent egg depending on the number of eggs already laid.

Following Martin et al.'s (2025) notation, our model for the observed clutch size at measurement t of female i was given by

$$\begin{aligned} \text{clutch size}_{ti} &\sim \text{ordered logit}(\mu_{ti}, \sigma_i, \theta) \\ \text{logit}(\mu_{ti}) &= \mu_{0i} + y_{0t} + (\beta_1 + \beta_{1i}) \text{date}_{ti} \\ &+ (\beta_2 + \beta_{2i}) \text{attempt}_{ti} + (\beta_3 + \beta_{3i}) \text{date} \times \text{attempt}_{ti} \\ \log(\sigma_i) &= \sigma_0 + \sigma_{0i} \\ [\boldsymbol{\mu}_0^\top, \boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top, \boldsymbol{\beta}_3^\top, \boldsymbol{\sigma}_0^\top]^\top &\sim \text{MVN}(\mathbf{0}, \mathbf{P}) \\ \mathbf{y}_0 &\sim N(\mathbf{0}, \sigma_y) \end{aligned} \quad (1)$$

where $^\top$ indicates transposition. As in a standard GLMM, the ordinal model is composed of expectation μ and dispersion σ parameters that govern the mean and residual variability, respectively, of measurements on the latent transformed scale. The latent expectation for a given measurement was determined by the individual reaction norm intercept μ_{0i} , population-level β_{1-3} and individual-level β_{1i-3i} slopes in response to the laying date standardized (mean = 0, variance = 1) within each population, her laying attempt order (also standardized), and their interaction, as well as a random intercept for the year y_{0t} during which a measurement was taken. The dispersion was determined by population-level (σ_0) and individual-level (σ_{0i}) reaction norm residual parameters, which quantified heterogeneity among females in residual clutch size variation, which is determined by the total value of σ . Note that we use bold font to distinguish vectors of individual deviations $\boldsymbol{\mu}_0, \boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \boldsymbol{\beta}_3, \boldsymbol{\sigma}_0$ from the corresponding population average parameters, with the subscript i used to index specific individual values in these vectors. The covariance among these individual-level reaction norm parameters was estimated by the $\mathbf{P}$ (co)variance matrix. Unlike in a linear regression, the ordinal model separates out $K - 1$ population-level intercepts θ (termed “thresholds” or “cut points”) for K observed clutch sizes, such that the probability of measuring a given clutch size k for observation clutch size ti is

$$\begin{aligned} P(\text{clutch size}_{ti} = k \mid \mu_{ti}, \sigma_i, \theta_k) &= \Lambda\left(\frac{\theta_k - \mu_{ti}}{\sigma_i}\right) \\ &- \Lambda\left(\frac{\theta_{k-1} - \mu_{ti}}{\sigma_i}\right). \end{aligned} \quad (2)$$

Here, θ_k is the threshold for a given clutch size k , and Λ indicates the logistic cumulative distribution function. Fortunately, interpretations of the expectation μ and dispersion σ parameters remain intuitive as in a linear model, with higher μ increasing the probability of crossing the thresh-

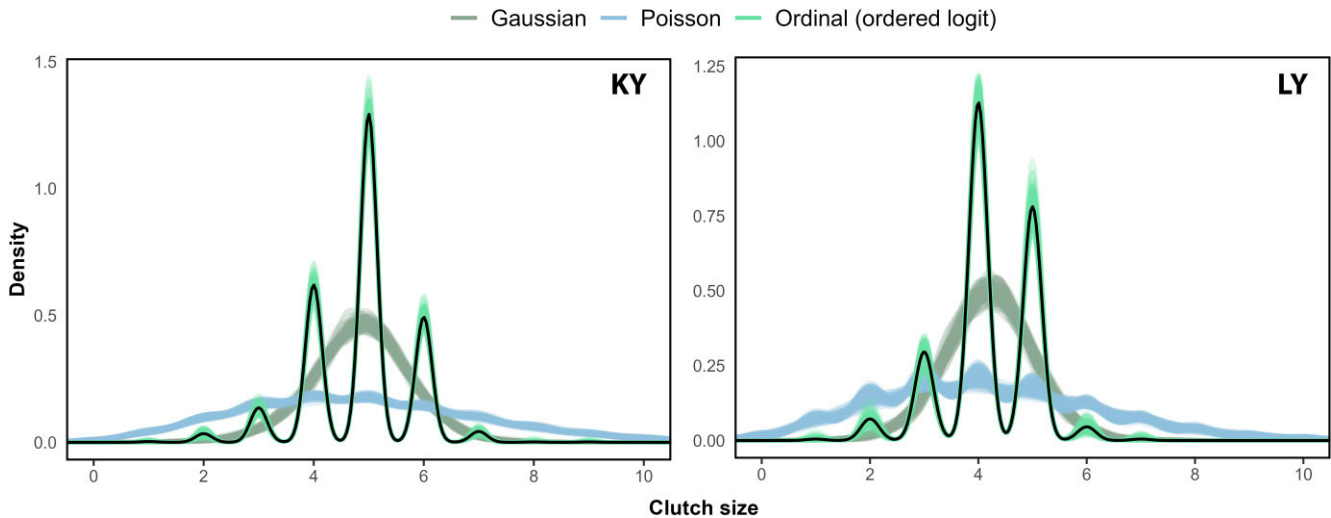


Figure 1. Posterior predictive check of house sparrow clutch sizes for Kentucky (KY) and Lundy (LY) samples. Density estimates (y-axis) are shown for the distribution of clutch sizes (x-axis) and predictions from Gaussian (gray), Poisson (blue), and ordinal/ordered logit (green) models of this data. Predicted densities are shown for 100 random MCMC samples from the posterior distributions of each model, which overlap to form a band of predictive uncertainty. The performance of each model can be visually assessed by comparing its predicted densities to the true density distribution of the raw data (black). Underdispersion in the data can be seen by the fact that clutch sizes of four to six have much higher density than predicted by Gaussian and Poisson models. The ordinal model captures this underdispersion and thus does a much better job at predicting the distribution of measurements.

olds θ , such that larger clutch sizes are more likely to be observed, and higher σ increasing the spread of probabilities across thresholds, such that greater variability is expected in observed clutch sizes. Therefore, females with positive μ_{0i} had higher than average mean clutch sizes, while females with positive σ_{0i} had higher than average variance (spread of probabilities) in their clutch sizes across measurements. Selection on mean clutch sizes across environments should, therefore, manifest through selection on the individual reaction norm intercept μ_{0i} , while selection on the variability of clutch sizes within or across environments should manifest through selection on the individual reaction norm slopes β_{1i-3i} and/or residual parameter σ_{0i} .

We used the R package *brms* (Bürkner, 2017) to initially characterize among-female variation in reaction norms (Supplementary Table S1), but due to constraints in that package, we subsequently used the more biologically intuitive parameterization of clutch size residuals described by Equations 1 and 2 for selection analysis in Stan (Carpenter et al., 2017).

Fitness components

To examine putative tradeoffs and their effects on clutch size evolution, we partitioned fitness into distinct components capturing variation in the quantity and quality of offspring. Offspring quantity was estimated as the minimal number of hatchlings produced from a female's clutch (n_{hatch}), while quality was assessed by the probability of survival (p_{survive}), based on the number of hatchlings surviving (n_{survive}) out of the total number of hatchlings (n_{hatch}), as well as average nestling mass (m_{nestling}). We assumed that variation in these fitness components was determined by maternal fitness rather than nestling phenotypes, motivating our treatment of offspring survival as a component of maternal fitness (Wolf & Wade, 2001). A female's total fitness was, therefore, estimated as total brood mass from the multiplicative effect of her fitness

components,

$$W_i = n_{\text{hatch},i} p_{\text{survive},i} m_{\text{nestling},i} \quad (3)$$

We estimated (non)linear selection effects caused by clutch size reaction norms on each fitness component and then subsequently combined selection gradients across models, following McGlothlin (2010), to estimate the total magnitude of nonlinear selection expected because of putative tradeoffs.

To parsimoniously reduce model complexity and enhance power, we only included reaction norm parameters in the final (non)linear selection model for which we had evidence of biologically meaningful individual differences (among-individual variance > 0). See Supplementary Table S2 for regression coefficients from a full model, including all parameters. Therefore, we simply denote the final set of reaction norm parameters for female i considered in the selection analysis by $\mathbf{z}_i$ to describe the fitness component models (see results for further details). The formal structure of covariates and random effects was constant across fitness components, so we use a generalized notation following Martin et al. (2025) to describe these models:

$$\begin{aligned} \text{fitness component}_{ti} &\sim f(\mu_{ti}, \sigma, \dots) \\ g_{\mu}(\mu_{ti}) &= \mu_0 + c_{0i} + y_{0t} + \beta_1 \text{date}_{ti} + \beta_2 \text{attempt}_{ti} \\ &+ \beta_3 \text{date} \times \text{attempt}_{ti} + \mathbf{b}^T \mathbf{z}_i + \mathbf{z}_i^T \mathbf{Q} \mathbf{z}_i \\ g_{\sigma}(\sigma) &= \sigma_0 \\ c_{0i} &\sim N(0, \sigma_c) \\ y_{0t} &\sim N(0, \sigma_y) \end{aligned} \quad (4)$$

Here, fitness component $_{ti}$ is either the hatchling count n_{hatch} , the number of surviving nestlings n_{survive} , used to calculate the probability of nestling survival p_{survive} given n_{hatch} , or average nestling mass m_{brood} for female i at obser-

vation t , which is described by a probability distribution f with link functions g on the expectation μ and dispersion σ of measurements. These measures were analyzed together in a multi-response/multivariate model structure, with distinct link functions used based on the distribution assumed for each fitness component. For n_{hatch} , the distribution was an ordered logit model as with clutch size, including the additional intercept threshold parameters θ ; for p_{survive} , the distribution was a binomial, such that μ is the expected probability of survival estimated using the count of successful survival n_{survive} out of total hatchlings n_{hatch} , with $\text{logit}(p_{\text{survive}})$ being the log-odds of survival, and σ is Gaussian noise on the latent scale (i.e., an observation-level random effect) capturing overdispersion with respect to the expected binomial variance (Harrison, 2014). Mean-scaled m_{brood} was modeled with a normal distribution. Note that model predictions for mass were observed in the range of proper non-negative values.

As in the clutch size model, $\beta_1 - \beta_3$ accounted for the average effects of a female's within-season laying date and its interaction with her laying attempt on her fitness components, with the random effect y_0 capturing any unmeasured year-specific effects. Accounting for the population-level fixed effects in the fitness model was important to adjust for expected variation in fitness as a function of laying attempt and order, which may confound direct measurement of selection on among-female variation in clutch size reaction norms. Note that exclusion of these effects does not meaningfully alter estimated patterns of selection (Supplementary Table S3). Regression coefficients $\mathbf{b}$ described the main effects of each modeled clutch size reaction norm parameter, including the female-specific residual variation σ_{0i} , on the fitness component, while $\mathbf{Q}$ was a symmetric matrix containing regression coefficients $\mathbf{q}$, with the diagonal coefficients capturing nonlinear effects for the same reaction norm parameter (e.g., intercept $\times$ intercept) and the off-diagonal coefficients capturing nonlinear effects among distinct reaction norm parameters (e.g., correlational selection on intercepts and residuals). For instance, if $\mathbf{z}_i = [\mu_{0i}, \sigma_{0i}]^T$ contains the clutch size reaction norm intercept and residual variation of female i , $\mathbf{b}^T \mathbf{z}_i = b_1 \mu_{0i} + b_2 \sigma_{0i}$ and $\mathbf{z}_i^T \mathbf{Q} \mathbf{z}_i = q_1 \mu_{0i}^2 + q_2 \sigma_{0i}^2 + q_3 \mu_{0i} \sigma_{0i}$. To directly test Lack's (1947) predictions, we also examined whether laying date modified the strength of these selection effects on clutch size reaction norms within seasons. Finally, the random effect c_{0i} quantified any repeatable, individual-level effects on the fitness component unexplained by the selection coefficients and thus represented putative sources of unexplained selection (Martin et al., 2025).

Our full (non)linear reaction norm selection model estimated Equations 1 and 4 simultaneously as a single multi-response/multivariate Bayesian GLMM, including clutch size and all three fitness components in a single analysis. As suggested by Martin et al. (2025), we applied weakly regularizing priors for all model parameters to facilitate more robust inference. The model was estimated in the Stan probabilistic programming language (Carpenter et al., 2017) using brms (Bürkner, 2017) as well as the cmdstan package (Gabry et al., 2023) in R (R Core Team, 2025), with appropriate convergence ensured by Rhat values < 1.01 , the absence of divergent transitions, and appropriately large bulk and tail effective sample sizes (Gelman et al., 2020). Posterior estimates of selection effects $\mathbf{b}$ and $\mathbf{q}$ from this multi-response GLMM were then subsequently transformed us-

ing partial derivatives to calculate standardized directional β_{z_i} and quadratic γ_{z_i} selection gradients with respect to the original scale of our data (Lande & Arnold, 1983; Martin et al., 2025; Morrissey & Sakrejsda, 2013), while still retaining statistical uncertainty in the process (Stinchcombe et al., 2014). Note that this step included the doubling of the off-diagonal elements in $\mathbf{Q}$ that is necessary for calculating γ_{z_i} (Stinchcombe et al., 2008), as well as the mean-scaling of the relationship between fitness and phenotype. These gradients allowed us to interpret patterns of directional and quadratic (stabilizing/disruptive and correlational) selection on reaction norm parameters using standardized effect sizes, facilitating comparison across the distinct link functions used for the reaction norm parameters and fitness components.

Our Bayesian analysis provided estimates of posterior distributions for each model parameter and quantity generated from them, which were subsequently used to quantify uncertainty for all effect sizes of interest as well as to conduct hypothesis tests for directional effects. We summarize these distributions below using posterior medians as a robust estimate of expected effect size, 90% Bayesian credible intervals (CIs) as measures of statistical uncertainty, as well as posterior probabilities as measures of support for directional effects (i.e., the probability in support of a positive $p+$ or negative $p-$ selection effect). We use 90% Bayesian CIs because of their intuitive interpretation with respect to traditional evidential standards for hypothesis testing. A 90% Bayesian CI excluding zero implies a posterior probability $> .95$ for the direction of the estimated effect, and thus a probability $< .05$ against this effect (Olivier et al., 2024). However, it is important to emphasize that, despite the common tendency to try to recapitulate null hypothesis testing within a Bayesian framework, Bayesian CIs often do not conform to classical confidence intervals (Nalborczyk et al., 2019) and should not be interpreted equivalently, nor used for dichotomous interpretations of statistical "significance," the dangers of which are well established (McShane et al., 2019). For Bayesian inference, the choice of CI is fundamentally arbitrary with respect to the entire posterior distribution, and the strength of evidence for directional effects should always be interpreted continuously with respect to the conservatism of priors, the complexity of the estimated model, and the distribution of the expected range of biologically meaningful effect sizes (McElreath, 2020).

Results

Reaction norms of clutch size and fitness components

Despite using a distinct probability distribution, we found that clutch sizes exhibited the same complex, qualitative patterns in average reaction norms as found in earlier analyses (Supplementary Table S1, Figure 2A; Westneat et al., 2009, 2014). Clutch size declined with date in the season (see Supplementary Table S1) and increased with attempt order but also declined faster with date at higher attempt orders. Absolute parameter values for reaction norms were very similar across sites, although Lundy exhibited a more positive clutch size slope with respect to lay attempt ($\Delta\beta_2 = 0.15$, 90% CI = [0.06, 0.24], $p_+ = .99$). The fitness components also responded in turn to differences in females' lay-

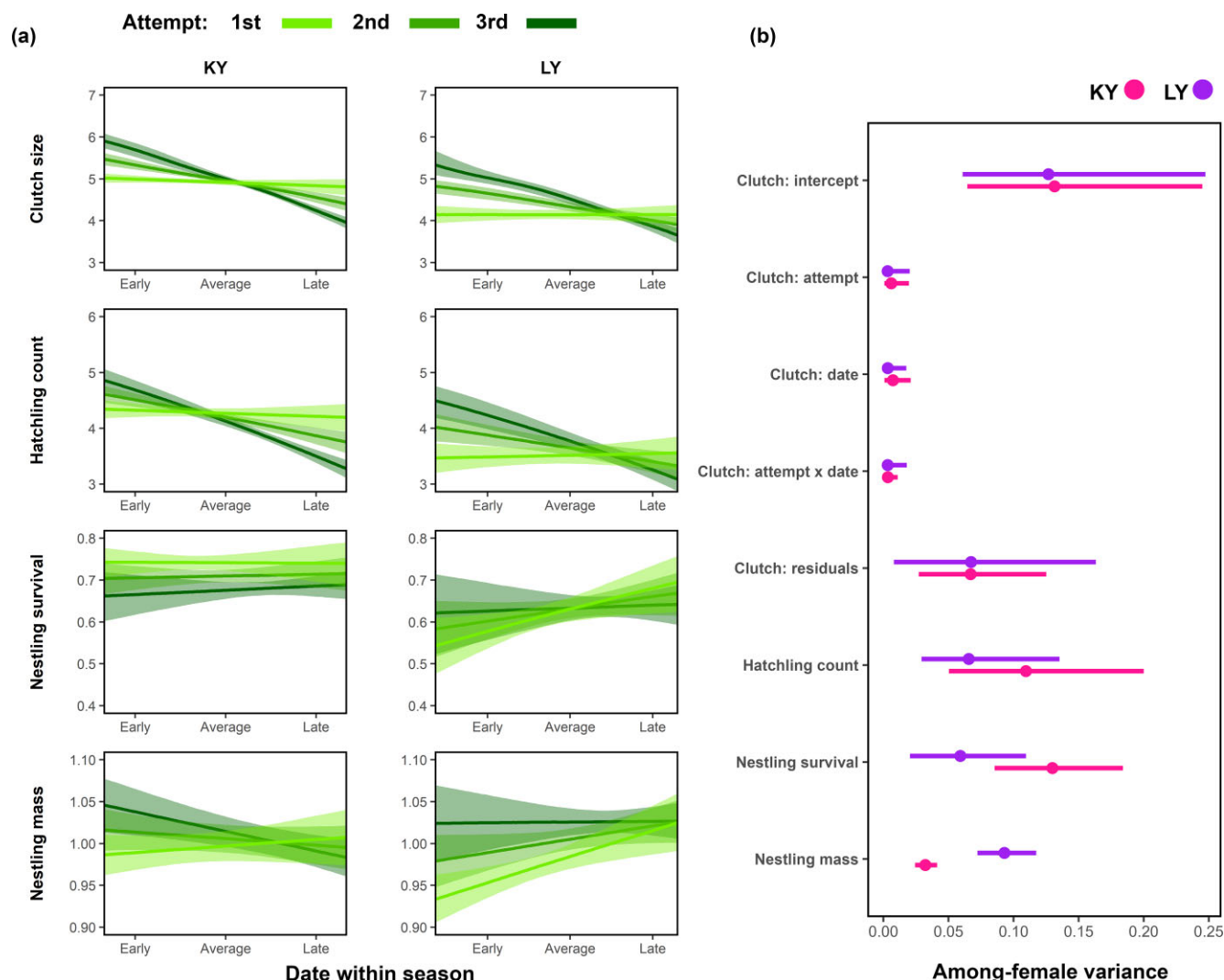


Figure 2. Clutch size and fitness component reaction norms from two populations of house sparrows. (A) The left panel shows population reaction norms for the Kentucky (KY) and Lundy (LY) populations describing how a female's relative laying date within the season (early = -2 SD, average = mean date, late = $+2$ SD) and the order of her attempt (first–third attempt) affect her clutch size, hatchling count, proportion of survival nestlings, and expected nestling mass. See [Supplementary Figure S2](#) for raw clutch size distributions across attempt orders. Subsequent selection analyses were adjusted for these average effects. (B) The right panel shows the magnitude of variance among females due to repeatable individual differences in clutch size reaction norms and fitness components (KY = pink, LY = purple). The Gaussian nestling mass variance was standardized by the residual SD for comparison. Expected effect sizes (posterior medians) are indicated by dark lines (left) or enlarged center points (right), with 90% posterior credible intervals indicated by the surrounding bands.

ing date and attempt order ([Supplementary Table S1](#), [Figure 2A](#)). Hatchling count decreased with date in the season and increased with increasing attempt order, while the main effect of attempt order on hatchling count differed between study sites. Nestling survival declined with attempt order in Kentucky but did not clearly differ as a function of date or its interaction with attempt, while moderate evidence was found for an interaction between date and attempt in Lundy. In both populations, nestlings were heavier on average across lay attempts, and mass increased by date for first broods, but this relationship declined at higher attempt orders ([Figure 2A](#)). Despite small differences, average reaction norms were highly similar across populations.

Among-individual variance

Repeatable individual differences were also observed in the intercept and residual parameters of females'

clutch size reaction norms in both populations ([Figure 2B](#)), indicating that females differed both in their mean clutch sizes (KY: $\text{var}(\mu_0) = 0.13$, 90% CI [0.06, 0.25]; LY: $\text{var}(\mu_0) = 0.13$, 90% CI [0.06, 0.25]) as well as the variability of their clutch sizes across breeding attempts (KY: $\text{var}(\sigma_0) = 0.07$, 90% CI [0.03, 0.13]; LY: $\text{var}(\mu_0) = 0.07$, 90% CI [0.01, 0.16]). [Supplementary Figure S1](#) shows the variation estimated in expected female clutch size distributions. However, individual differences in reaction norm slopes for date (KY: $\text{var}(\beta_1) = 0.01$, 90% CI [0.00, 0.02]; LY: $\text{var}(\beta_1) = 0.00$, 90% CI [0.00, 0.02]), attempt order (KY: $\text{var}(\beta_2) = 0.01$, 90% CI [0.00, 0.02]; LY: $\text{var}(\beta_2) = 0.00$, 90% CI [0.00, 0.02]), and the date $\times$ attempt interaction (KY: $\text{var}(\beta_3) = 0.00$, 90% CI [0.00, 0.01]; LY: $\text{var}(\beta_3) = 0.00$, 90% CI [0.00, 0.02]) were small and near to zero, suggesting that slopes in response to these environ-

mental cues were relatively fixed in both populations. This small magnitude of among-female variance provided us with limited scope to detect putative selection on these responses, and we therefore excluded the reaction norm slopes from subsequent selection analysis. Overall, patterns of among-individual variance in clutch size reaction norms were similar across Kentucky and Lundy. In addition, we observed repeatable individual differences across the fitness components (Figure 2B), with modest among-female variation in hatchling count (KY: $\text{var}(c_0) = 0.11$, 90% CI [0.05, 0.20]; LY: $\text{var}(c_0) = 0.07$, 90% CI [0.03, 0.14]), nestling survival (KY: $\text{var}(c_0) = 0.13$, 90% CI [0.09, 0.18]; LY: $\text{var}(c_0) = 0.06$, 90% CI [0.02, 0.11]), and average nestling mass (KY: $\text{var}(c_0) = 0.03$, 90% CI [0.02, 0.04]; LY: $\text{var}(c_0) = 0.09$, 90% CI [0.07, 0.12]). This indicated potential for ongoing selection of these fitness components to shape clutch size reaction norms.

Multivariate analysis of selection

Variation in nestling survival did not result in selection on any clutch size reaction norm parameters. However, we found clear evidence of tradeoffs between offspring quantity and quality due to conflicting patterns of selection via hatchling counts and expected nestling mass (Table 1, Figure 3). In both populations, directional selection via hatchling count acted to increase mean clutch size through the reaction norm intercept, as well as to decrease clutch size variability through the reaction norm residual parameter. In Kentucky, stabilizing selection occurred via hatchling count on clutch size reaction norm intercepts. No clear evidence was found for correlational selection between intercepts and residuals (Table 1). This resulted in a skewed hill-shaped selection surface (Figure 3B). In contrast, directional selection via offspring mass favored smaller mean clutch sizes and higher variability, with stabilizing selection on residuals, though directional selection was not observed on clutch size intercepts via nestling mass on Lundy. This resulted in a saddle-shaped selection surface in Kentucky, with a more hill-shaped surface in Lundy (Figure 3C).

Combining selection gradients across fitness components, we found that the observed quantity–quality tradeoff weakened total directional selection on clutch size intercepts (KY: $\beta_{\mu_0} = 0.11$, 90% CI [0.08, 0.14]; LY: $\beta_{\mu_0} = 0.12$, 90% CI [0.08, 0.16]) and residuals (KY: $\beta_{\sigma_0} = 0.04$, 90% CI [0.00, 0.06]; LY: $\beta_{\sigma_0} = 0.05$, 90% CI [−0.00, 0.05]), while also increasing the strength of stabilizing selection on female-specific residual variation (KY: $\gamma_{\sigma_0} = -0.08$, 90% CI [−0.15, −0.03]; LY: $\gamma_{\sigma_0} = -0.13$, 90% CI [−0.21, −0.05]) and diminishing the stabilizing selection on intercepts via hatchling count observed in Kentucky ($\gamma_{\mu_0} = -0.01$, 90% CI [−0.06, 0.04]). See Figure 3D for a visualization of the combined selection surfaces.

Laying date effects on the selection of clutch size

Finally, we also found evidence that selection gradients fluctuated within season as a function of laying date. We estimated these fluctuations by quantifying interaction effects of date on the b and q coefficients in the selection model (Equation 4), denoted herein Δb and Δq . In KY, the strength of directional selection on clutch size reaction norm intercepts via hatchling count decreased later in the season

($\Delta b_{\mu_0 \times \text{date}} = -0.43$, 90% CI [−0.71, −0.23]). An effect of this magnitude is expected to greatly dampen but not reverse the direction of selection (Supplementary Table S2). In turn, stabilizing selection on female-specific residual variation via nestling mass increased at later laying dates, as indicated by the increasingly negative quadratic regression coefficient ($\Delta q_{\sigma_0 \times \text{date}} = -1.61$, 90% CI [−2.95, −0.35]). In LY, the relative strength of stabilizing selection ($\Delta q_{\sigma_0 \times \text{date}} = -1.53$, 90% CI [−2.95, −0.23]) via nestling mass on reaction norm residuals also increased at later laying dates. See Supplementary Table S4 for full results.

Discussion

Our analysis of nonlinear selection on clutch size reaction norms revealed several intriguing results relevant to the evolution of phenotypic plasticity, confirming but also extending traditional models of the quantity–quality tradeoffs shaping females’ reproductive timing and tempo. Consistent with Lack’s classic hypothesis, we observed opposing patterns of selection acting to constrain the evolution of expected female clutch sizes. Building on prior findings (e.g., Perrins, 1965; Pettifor et al., 1988; Slagsvold & Lifjeld, 1988), we observed in both Kentucky and Lundy that selection via hatchling count favored larger intercept clutch sizes (clutch size at the mean laying date and attempt, Figure 3B). In contrast to Lack’s hypothesis, this is not always linear, as we observed some evidence of diminishing hatching success for large clutch sizes in Kentucky. This was not sufficient to reverse directional selection via hatchling number, but we also found that selection via nestling mass favored lower intercepts (Figure 3C). This supports the key idea of Lack’s hypothesis: more eggs produce more offspring, but their quality is reduced by their quantity, favoring lower mean clutch sizes. This tradeoff among components is expected to result in stabilizing selection around an optimal clutch size, as Lack proposed (Figure 3D). The qualitative direction of our results did not bear out quantitatively, as the combined selection estimates indicated that quantity was exerting a stronger average effect on females’ total fitness and clutch size was not at an optimum. However, it is likely that nestling mass at fledging has downstream effects on post-fledgling survival (e.g., Schwagmeyer & Mock, 2008) that were unobserved in our study, such that the total magnitude of stabilizing selection due to quantity–quality tradeoffs is likely underestimated. Nevertheless, several aspects of our analysis provide new insights not only for understanding the evolution of clutch size in birds but also the ways that selection might act on reaction norms more generally.

We could not measure selection on females’ clutch size slopes due to the absence of substantive among-individual variance (Figure 1). Given our relatively large sample size, our study should be sufficiently powered to detect small variation in slopes, which suggests that most females are exhibiting very similar patterns of plasticity toward date within a breeding season and the order of their laying attempt. These plastic changes were not only relatively fixed within populations but also exhibited similar magnitude and direction across populations (Supplementary Table S1). Our findings are, therefore, consistent with a history of persistent selection on females’ date and attempt order slopes, such that the reaction norms of both sparrow populations now exhibit nearly uniform patterns of multidimensional

Table 1. Standardized selection gradients (directional, β , and quadratic, γ) on clutch size reaction norm individual intercepts (μ_0), individual residual variations (σ_0), and their correlation ($\mu_0\sigma_0$) via three fitness components from two populations of house sparrows.

Standardized selection gradient	Kentucky			Lundy Island		
	Median	90% CI	<i>p</i>	Median	90% CI	<i>p</i>
Hatchling count						
β_{μ_0}	0.14	0.12, 0.16	1.00	0.13	0.11, 0.16	1.00
β_{σ_0}	−0.03	−0.05, −0.02	.99	−0.04	−0.07, −0.01	.99
γ_{μ_0}	−0.04	−0.07, −0.01	.99	0.01	−0.04, 0.06	.61
γ_{σ_0}	−0.02	−0.05, 0.01	.84	−0.01	−0.06, 0.04	.61
$\gamma_{\mu_0\sigma_0}$	0.00	−0.01, 0.02	.64	0.01	−0.02, 0.03	.69
Nestling survival						
β_{μ_0}	0.01	−0.01, 0.02	.70	0.00	−0.03, 0.02	.55
β_{σ_0}	0.00	−0.01, 0.02	.58	0.02	−0.01, 0.04	.88
γ_{μ_0}	0.00	−0.03, 0.03	.51	0.00	−0.05, 0.04	.57
γ_{σ_0}	0.00	−0.03, 0.01	.67	−0.02	−0.07, 0.01	.85
$\gamma_{\mu_0\sigma_0}$	0.00	−0.02, 0.01	.54	0.00	−0.02, 0.01	.64
Nestling mass						
β_{μ_0}	−0.03	−0.06, −0.00	.96	−0.01	−0.04, 0.02	.79
β_{σ_0}	0.06	0.05, 0.08	1.00	0.07	0.04, 0.09	1.00
γ_{μ_0}	0.01	−0.02, 0.04	.79	−0.02	−0.05, 0.01	.80
γ_{σ_0}	−0.05	−0.10, −0.03	1.00	−0.1	−0.15, −0.06	1.00
$\gamma_{\mu_0\sigma_0}$	0.00	−0.03, 0.03	.63	−0.01	−0.05, 0.01	.80

Note. Median estimates are reported for each effect size with accompanying 90% Bayesian credible intervals (CIs) of their posterior distributions and posterior probabilities (*p*) in the direction of the median (a value closer to 1 indicates stronger support for a directional effect, with $1 - p$ indicating support for an effect in the opposite direction). Gradients are standardized by the SD of among-female differences in each parameter.

plasticity with respect to clutch order and date. Under such a scenario, we expect that selection may have “erased its traces” (Haller & Hendry, 2014) through its own efficacy in shaping clutch size plasticity, such that it can no longer be straightforwardly detected in contemporary populations, challenging our ability to use naturally occurring variation to test whether Lack’s hypothesis also exerts stabilizing selection on reaction norm slopes.

Nonetheless, independently of systematic changes in their response to date within season, females exhibited pronounced individual differences in residual variation—that is, the variation in which their clutch sizes deviated from their expected reaction norm clutch size, which may reflect the aggregated consequences of various other unmeasured reaction norm slopes (Martin et al., 2025; Westneat et al., 2015). This allowed us to observe whether deviations from optimal mean clutch sizes were selected for or against across fitness components, testing more nuanced models of how putative tradeoffs and nonlinear adaptive landscapes may be shaping the entire phenotypic distribution of clutch sizes. Intriguingly, we also found evidence of contrasting selection effects across fitness components acting to constrain clutch size variability, but in the opposite direction to the pattern observed for mean clutch size. A reduced residual variance in clutch size was favored through hatchling count, but a larger clutch size residual variance was favored via nestling mass. Stabilizing selection (negative quadratic selection) also occurred on reaction norm residuals, suggesting that the currently observed level of clutch size variability is likely near to a local optimum in both populations.

How this works is not completely clear, but if we consider selection gradients together across reaction norm intercepts and residuals, our results suggest that contemporary variation in females’ clutch size in both populations may be occurring near to an asymmetric fitness ridge, such that trade-offs are acting through opposing patterns of nonlinear selection on clutch size mean (intercepts) and variability (residuals) with respect to each population’s position on the ridge.

Jensen’s inequality, which describes how variance in inputs (phenotypes) reduces/increases outputs (fitness) from decelerating/accelerating functions (Ruel & Ayres, 1999), is often invoked to explain the evolution of apparently suboptimal trait values near to fitness ridges or cliffs. When there are greater costs to overshooting versus undershooting the fitness peak (e.g., because most offspring perform very poorly if there are too many and the environment is also poor; Mathot et al., 2017), the phenotypic mean is likely to evolve away from the peak of the fitness ridge to reduce the detrimental effects of naturally occurring variability in the phenotype (Martin & Huey, 2008; Vercken et al., 2012). However, most theoretical models of asymmetric ridges and cliffs have assumed an exogenously fixed level of phenotypic variability, ignoring the fact that selection can also shape reaction norm residuals and thus the magnitude of deviations from the mean phenotype.

Allowing for the evolution of phenotypic variability (or canalization) opens new pathways for population movement on the adaptive landscape. We expect that fitness can be maximized in proximity to such an asymmetric fitness ridge by either increasing the mean while reducing variability, or decreasing the mean while increasing variability, depending on the slope of the relationship between fitness and phenotype and how it is shaped by stochastic environmental factors (Figure 4). While greater and more canalized reproductive output can increase the expected rate at which a genotype proliferates, all else being equal, variation in offspring survival due to changes in environmental quality will also reduce this rate, selecting for strategies that buffer against and take advantage of environmental variation through responses such as reduced reproductive output and increased investment in offspring (León, 1983; Pásztor et al., 2000). In such cases, being conservative may be less costly if unexpectedly good conditions lead to faster development and high body mass of offspring. This is consistent with the contrasting patterns of selection detected in our study, with hatchling count favoring larger and less variable clutch sizes

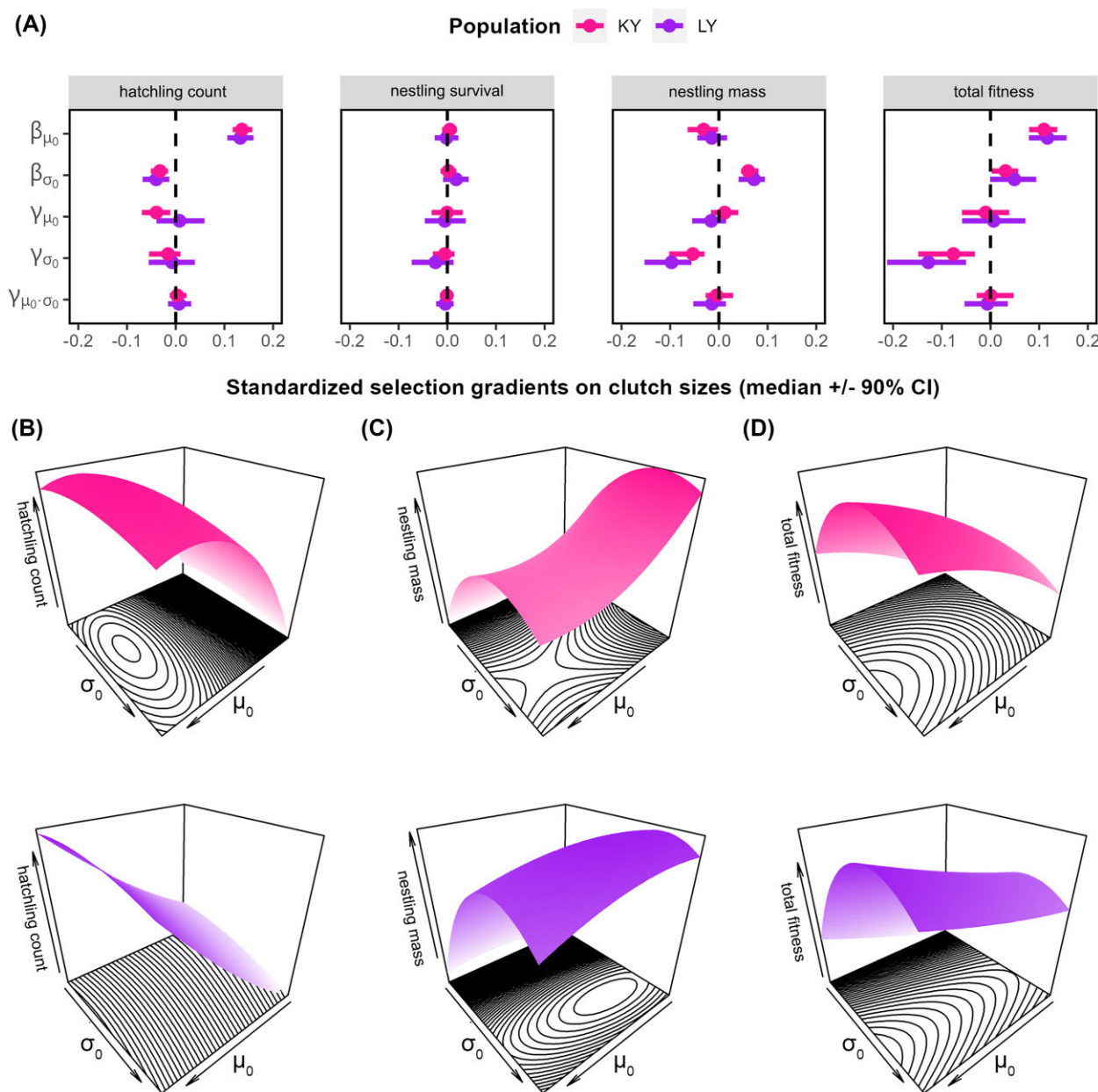


Figure 3. Component-specific and total selection gradients on clutch size reaction norm intercept μ_0 and residual variation σ_0 parameters in two populations of house sparrows (KY = pink, LY = purple). The μ_0 parameter increases the mean clutch size produced while the σ_0 parameter shapes the variability of clutch sizes around the mean (see Equations 1 and 2). (A) Standardized estimates of directional and quadratic gradients are summarized in the top panel by their posterior median (dot) and 90% Bayesian CI (lines). (B–D) Expected selection surfaces combining median selection effects for hatchling count (B), nestling mass (C), and combined fitness (D; see Equation 3). The bottom x-axis and y-axis show the magnitudes of reaction norm intercepts and residual variation (arrows indicate increasing magnitude, i.e., higher clutch size means and higher variances, respectively), while the vertical z-axis shows expected relative fitness with respect to the fitness components and combined total fitness. The curvature and height of the surface approximate the local adaptive landscape with respect to these parameters, with an accompanying contour mapping shown below.

and nestling mass favoring sm and more variable clutch sizes. Given these considerations, understanding whether plasticity in sparrows' parental investment mediates the observed selection effects via nestling mass is an important direction for future research.

Taken together, our results suggest that house sparrows face a tradeoff between two differently shaped clutch size *distributions*, each of which better maximizes fitness with respect to either offspring quality or quantity (Figure 4). This tradeoff is expected to constrain directional selection

on clutch size intercepts and residuals while also increasing the magnitude of stabilizing selection on clutch size variability, suggesting that the average magnitude of clutch size variability in both populations may be nearly optimal with respect to total fitness, despite being potentially suboptimal with respect to hatchling count or nestling mass considered in isolation. Our results, therefore, add important and rare empirical data to the large body of theory predicting that asymmetries in the adaptive landscape are likely to be ubiquitous and key to understanding the evolution of

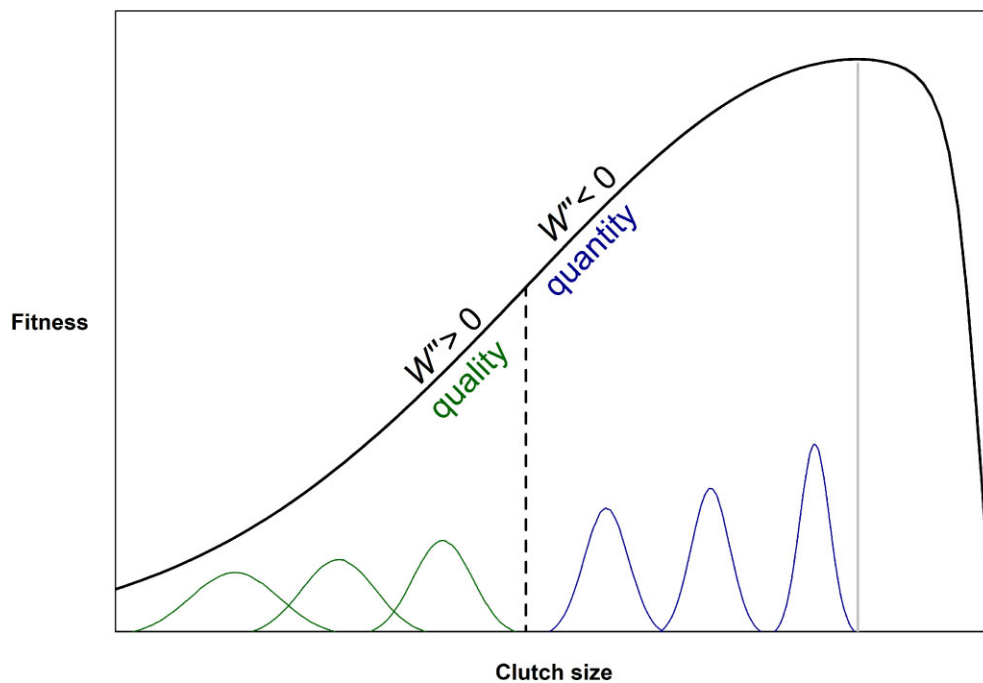


Figure 4. A hypothetical cliff-shaped fitness function is shown, based on [Martin and Huey \(2008\)](#), where fitness steadily increases and then rapidly decreases with clutch size (e.g., due to excessive energetic demands and the inability to share parental care among more offspring). The asymmetry of the fitness ridge causes positive deviations from the optimum to more severely reduce fitness than negative deviations. The gray line indicates the optimum of the fitness function, where selection would be expected to push the population in the absence of phenotypic variability. However, unpredictable environmental effects and the potential costs of canalization make some degree of variability in clutch size inevitable, as indicated by the hypothetical phenotypic distributions at the bottom of the graph. The magnitude of these effects and the degree to which selection can shape clutch size variability will determine where the phenotypic distribution is expected to evolve with respect to the ridge. If females can canalize their reproductive output to environmental perturbations across seasons, selection for offspring quantity can push the mean clutch size closer toward the fitness optimum while also reducing variability in clutch sizes, with greater selection against variability expected with greater deceleration of the slope of the fitness function (i.e., as the second derivative $W'' < 0$ becomes more negative). However, offspring quantity is expected to trade off with offspring quality near to the fitness optimum, due to females' inability to canalize their investment in response to variable environmental quality. Therefore, selection for offspring quality may instead favor a reduced mean clutch size lower on the slope of the fitness function, as well as greater variability in clutch sizes, with females plastically regulating their reproductive output, taking advantage of and buffering against stochastic fluctuations in their environment. The strength of selection for variability is expected to increase with greater acceleration of the slope of the fitness function (i.e., $W'' > 0$).

(sub)optimal trait values in the wild ([León, 1983](#); [Martin & Huey, 2008](#); [Pásztor et al., 2000](#); [Ruel & Ayres, 1999](#); [Vercken et al., 2012](#)). Future theoretical models and empirical studies should incorporate a more explicit reaction norm perspective, as we have done here using [Martin et al.'s \(2025\)](#) statistical approach, to better understand how multivariate selection affects plasticity and the shape of phenotypic distributions in response to such complex and asymmetric fitness functions.

Supplementary material

Supplementary material is available online at [Evolution](#).

Data availability

All data and code for replicating our analyses are archived and openly available on GitHub (<https://github.com/Jordan-Scott-Martin/NLSclutch>) and Zenodo at [Martin, J. S., Westneat, D., Nakagawa, S., Schroeder, J., & Burke, T. \(2025\)](#). Code and data from: Measuring selection on reaction norms: Lack's principle and plasticity in clutch size (v1.1). Zenodo. (<https://doi.org/10.5281/zenodo.16749053>).

Author contributions

J.S.M. and D.F.W. share first authorship. D.F.W. conceived the original idea, managed data collection in the Kentucky population, created the combined dataset, explored several possible statistical approaches, and wrote an early draft. J.S.M. solved many issues with the analysis, completely overhauled the approach, and rewrote the manuscript following the reanalysis. S.N. provided extensive statistical advice during early phases of this study and helped collect data from the Lundy Island population. S.N., J.S., and T.B. managed the data collection from Lundy Island. All authors contributed to the final draft of the manuscript.

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Conflict of interest

The authors declare that none of us has any interest or relationship, financial or otherwise that might be perceived as influencing an author's objectivity or considered a potential source of conflict of interest.

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